

Total No. of Questions :6]

SEAT No. :

P68

[Total No. of Pages : 2

APR-17/BE/Insem-78

B.E. (Information Technology)

GREEN IT :PRINCIPLES AND PRACTICES

(2012 Pattern) (Elective - IV(C))

Time : 1 Hour]

[Max. Marks :30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.***
- 2) Neat diagrams must be drawn wherever necessary.***
- 3) Figures to right indicate full marks.***
- 4) Use of calculator is allowed.***
- 5) Assume suitable data if necessary.***

Q1) a) What are P and C States of processor? How does processor C-States save energy? ***[5]***

b) Is information technology a problem or a solution to sustainable development ***[5]***

OR

Q2) a) Explain the life cycle of an electronic device. Also based on your view discuss the environmental impact of each stage ***[5]***

b) How software's impact the environment and the energy consumption of computing systems? ***[5]***

Q3) a) What key IT infrastructure is used in data centers ***[6]***

b) How modifiability and reusability metrics are measured ***[4]***

OR

Q4) a) What are the quality attributes used for developing sustainable software ***[6]***

b) How server farm and server virtualization helps in saving energy ***[4]***

P.T.O

- Q5)** a) How storage virtualization and cloud storage help in saving energy [5]
- b) What are challenges of next-Generation networks with respect to green communication [5]

OR

- Q6)** a) Why is there an increasing need for green data storage? [5]
- b) How energy-efficient networking is related to objectives of green IT [5]

